Parallel Simulation of Microplastic Dispersion Using Ocean Current Data

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I. Introduction

Microplastic pollution has spread across every ocean basin, and the usual way to study where it travels is Lagrangian particle tracking: virtual particles are released into a measured surface current field and carried along as passive tracers. The idea is simple, but the computation is not. Every particle must be advanced on its own at each timestep, and each step rebuilds the surrounding current velocity by interpolation in space and time, so the work grows with particle count, run length and the cost of the integration scheme. A basin-scale study with stable statistics needs millions of particles followed over months of simulated time, and that is where a serial program stops being practical. **Why this is a problem:** the cost pushes researchers into a trade-off they should not have to make. Either they cut the particle count until the accumulation patterns turn noisy, or they coarsen the current field until the small eddies that gather plastic together are smoothed out. Repeated runs and sensitivity studies become too expensive, and answers arrive too late to be useful to decision makers. **Problem Statement:** simulating microplastic dispersion from real ocean current data, at a particle count and resolution useful for environmental analysis, is too slow under serial execution. What is needed is a parallel framework that models particle-level transport driven by ocean current fields efficiently, so larger and finer studies fit within practical runtimes on ordinary hardware.

Most existing work on microplastic transport comes from oceanography and uses Lagrangian frameworks driven by ocean current reanalysis data. It has shown that passive-tracer advection reproduces real transport behaviour well, and has been used to locate the subtropical gyres where floating plastic collects. Their focus is physical accuracy. The advection step itself is usually run serially, or handed to a shared institutional cluster, and its computational behaviour is hardly ever reported. We rarely learn how this interpolation-heavy, memory-bound update behaves on several threads, and speedup and parallel efficiency almost never appear next to the dispersion results. **Research Gap:** what is missing is a parallel version of the advection kernel that is measured properly, compared against an equivalent serial baseline on the same data, and that still runs on an ordinary multi-core machine rather than a supercomputer.

Our framework treats microplastic particles as passive tracers moving with the surface current. Each particle's velocity comes from bilinear interpolation over the current grid together with linear interpolation between the two nearest time slices, and positions are advanced with a fixed-step Runge–Kutta integrator. Since particles do not interact within a timestep, the advection loop is parallelised over the particle array using OpenMP, with attention to cache behaviour and thread load balance. **Beneficiaries:** marine and environmental researchers who want faster turnaround on dispersion experiments, coastal agencies trying to locate accumulation and beaching zones, and policymakers deciding where cleanup effort is worth spending.

Novelty: we are not proposing a new ocean transport model. What is new is treating the dispersion simulation as a parallel computing problem in its own right. As far as we know, this is among the first attempts to pair current-driven Lagrangian transport with a parallel implementation whose scaling is actually measured rather than assumed, against a matched serial baseline using the same data and numerics. Because the code stays small and avoids heavy simulation frameworks, whatever speedup we see belongs to the advection kernel itself.

Evaluation: the framework is assessed in two ways. Computationally we compare the parallel and serial versions on runtime, speedup and parallel efficiency while varying particle count, grid resolution and thread count, which should show where memory bandwidth and interpolation cost limit scaling. Physically we check that the dispersion patterns look reasonable for current-driven transport and agree with the serial baseline.

Expected contributions: the project delivers: (1) a Lagrangian particle transport model driven by real ocean current data, using spatial and temporal interpolation of the velocity field; (2) a serial baseline together with a parallel implementation of the particle advection kernel; (3) a scalability study reporting speedup and parallel efficiency across particle counts, resolutions and core counts; and (4) dispersion maps showing likely accumulation zones.